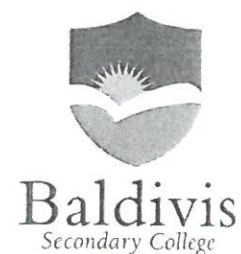


Name: Mark Scheme

Teacher: _____

Class: _____ Mark: _____ /25



YEAR 8
SEMESTER 1 2015
Algebra Test
Resources Allowed

Question 1

[3]

In the expression $7x^2 + 5x - 2$

(a) How many terms are there?

3 ✓

(b) What is the coefficient of x ?

5 ✓

(c) What is the constant term?

-2 ✓

Question 2

[2]

Philip thought of a number (n), he then doubles it and adds 10.

(a) Write an expression for this situation

$2n + 10$ ✓

(b) Find the value of this expression when n is equal to 12.

34 ✓

Question 3

[7]

Simplify the following:

(a) $2x + 8x = 10x$ ✓

(b) $5x + y - x - 4y$

$4x - 3y$ ✓

(c) $2m \times 8n$

$16mn$ ✓

(d) $5g \times 7g$

$35g^2$ ✓

END OF TEST

$$(e) \frac{5tu}{25uv}$$

$$\frac{t}{5v} \quad \checkmark$$

$$(f) \frac{6a^2b}{2ab^2}$$

$$\frac{3a}{b} \quad \checkmark$$

Question 4

[7]

Expand and simplify:

$$(a) 5(x + 3)$$

$$5x + 15 \quad \checkmark$$

$$(b) -2(y - 7)$$

$$-2y + 14 \quad \checkmark$$

$$(c) 2x(4x - 7y + 3)$$

$$8x^2 - 14xy + 6x \quad \checkmark$$

$$(d) 3(a + 2) + 3a$$

$$3a + 6 + 3a = 6a + 6 \quad \checkmark$$

$$(e) 5(x + 3) + 2(x + 6)$$

$$5x + 15 + 2x + 12 = 7x + 27 \quad \checkmark$$

Question 5

[6]

Factorise:

$$(a) 5x + 15$$

$$5(x + 3) \quad \checkmark$$

$$(b) 2x - 6$$

$$2(x - 3) \quad \checkmark$$

$$(c) 10y + 4z$$

$$2(5y + 2z) \quad \checkmark$$

$$(d) x^2 + 6x$$

$$x(x + 6) \quad \checkmark$$

$$(e) 4x^2 + 2x + 10$$

$$2(x^2 + x + 5) \quad \checkmark \checkmark$$

END OF TEST